

Solving Equations & Inequalities

Answer Key — Worked Solutions

MTH1W · Homework — Answer Key

Worked solutions with method lines. Curriculum: MTH1W (Grade 9 de-streamed math).

1. **[L1 · One-step]** Subtract 11: $x = 20 - 11 = 9$.
2. **[L1 · Two-step]** Add 6: $2x = 16$. Divide by 2: $x = 8$.
3. **[L1 · Both sides]** Subtract 3x: $2x = 8$. Divide by 2: $x = 4$.
4. **[L1 · Distribution]** Distribute: $4x - 8 = 12$. Add 8: $4x = 20$. $x = 5$.
5. **[L1 · Rearrange]** Subtract c from both sides: $x = y - c$.
6. **[L1 · Inequality]** Add 2: $x > 5$.
Graph: open circle at 5, arrow right.
7. **[L2 · One-step]** Multiply both sides by 6: $x = 42$.
8. **[L2 · Two-step]** Subtract 8: $-3x = -6$. Divide by -3: $x = 2$.
9. **[L2 · Both sides]** Subtract 5x: $4x - 4 = 12$. Add 4: $4x = 16$. $x = 4$.
10. **[L2 · Fractions]** Multiply by 5: $x - 3 = 20$. Add 3: $x = 23$.
11. **[L2 · Rearrange]** Subtract 2w: $2l = P - 2w$. Divide by 2: $l = (P - 2w)/2$.
12. **[L2 · Inequality]** Subtract 2: $3x \leq 12$. Divide by 3: $x \leq 4$.
Graph: closed circle at 4, arrow left.
13. **[L3 · Consecutive]** Let the integers be n and $n + 2$, so $n + (n + 2) = 46$.
 $2n + 2 = 46$, $2n = 44$, $n = 22$. The integers are 22 and 24.
14. **[L3 · Both-sides distribute]** Distribute: $5x - 5 = 3x + 9$. Subtract 3x: $2x - 5 = 9$.
Add 5: $2x = 14$. Divide by 2: $x = 7$.
15. **[L3 · Fractions]** Multiply every term by 4: $x + 2x = 36$.
 $3x = 36$, so $x = 12$.
16. **[L3 · Rearrange]** Subtract u: $v - u = at$. Divide by t: $a = (v - u)/t$.
17. **[L3 · Inequality (flip)]** Subtract 1: $-3x < 9$. Divide by -3 and reverse the sign: $x > -3$.
Graph: open circle at -3, arrow right.
18. **[L3 · Formula & evaluate]** Rearrange: $2A = bh$, so $h = 2A/b$.
 $h = 2(24)/6 = 48/6 = 8$.